

# Aman Asish Gupta

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## Education

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|---------------------------------------------|------------|
| <b>Birla Institute of Technology, Mesra</b> | Present    |
| M.Tech: Computer Science and Engineering    | CGPA: 9.32 |
| <b>Silicon University, Bhubaneswar</b>      | 2024       |
| B.Tech: Computer Science and Engineering    | CGPA: 9.00 |
| <b>Central Academy, Dadabari, Kota</b>      | 2019       |
| Intermediate (PCM)                          | 82%        |
| <b>D.A.V. Public School, Hazaribagh</b>     | 2017       |
| Matriculation                               | CGPA: 9.80 |

## Technical Skills

**Programming:** C, C++, Java, Python, SQL  
**Web & Backend:** HTML, CSS, JavaScript, Node.js, Express.js, REST APIs  
**Frameworks:** MERN Stack, Flutter  
**Databases:** MySQL, MongoDB, MongoDB Atlas  
**AI/ML:** Deep Learning, RAG (Fundamentals), OpenCV, Medical Image Analysis, Feature Extraction  
**Tools:** Git, GitHub, VS Code, MATLAB, Jupyter Notebook  
**Core Subjects:** Operating Systems, OOPs, DBMS, Computer Networks, DSA

## Projects

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                                           |              |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------|
| <b>Voice Commerce - Voice-Controlled E-Commerce</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | React, Node.js, Express, MongoDB, Groq AI | Live Demo    |
| <ul style="list-style-type: none"><li>Built a voice-driven shopping platform with voice-based product search, cart add/remove, checkout, and JWT-authenticated login, following an MVC architecture with CRUD REST APIs tested via Postman</li><li>Led the NLP module: built intent classification and entity extraction (product, quantity, size) using Groq AI, with custom prompt engineering and regex-based fallback for low-confidence or ambiguous voice commands</li><li>Maintained a Git workflow with a hosted main branch and a dev branch for integration, using separate feature branches for frontend, backend, and NLP development</li></ul> |                                           |              |
| <b>ElderNest - Health &amp; User Management System</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Node.js, Express, MongoDB Atlas, Flutter  | GitHub   APK |
| <ul style="list-style-type: none"><li>Developed a secure backend using Node.js + Express with encrypted authentication and REST APIs</li><li>Simulated real-time heart-rate monitoring using CSV with planned IoT integration (ESP32 + MAX30102)</li><li>Integrated MongoDB Atlas for scalable cloud storage and laid groundwork for Flutter dashboard and anomaly detection</li></ul>                                                                                                                                                                                                                                                                      |                                           |              |

## Experience

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|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------|
| <b>Birla Institute of Technology, Mesra</b>                                                                                                                                                                                                                                               | 2025 | Paper |
| Research: Image Processing & Enhancement                                                                                                                                                                                                                                                  |      |       |
| <ul style="list-style-type: none"><li>Developed a hybrid image-enhancement framework using wavelet preprocessing, entropy-guided filtering, NLM denoising, and diffusion refinement</li><li>Evaluated enhancement quality on BSD100 datasets using PSNR, SSIM, and AMBE metrics</li></ul> |      |       |
| <b>Indian Institute of Technology (BHU), Varanasi</b>                                                                                                                                                                                                                                     | 2024 |       |
| AI/ML Research Contributor                                                                                                                                                                                                                                                                |      |       |
| <ul style="list-style-type: none"><li>Worked on CNN-based medical image analysis for fracture and crack-feature extraction from bone X-ray images</li></ul>                                                                                                                               |      |       |
| <b>Syllogistek Systems Private Ltd.</b>                                                                                                                                                                                                                                                   | 2022 |       |
| Intern - MERN Stack                                                                                                                                                                                                                                                                       |      |       |
| <ul style="list-style-type: none"><li>Developed and tested full-stack web modules using HTML, CSS, JavaScript, ExpressJS, NodeJS, and MongoDB</li></ul>                                                                                                                                   |      |       |

## Achievements

- Secured AIR 5226 in GATE (CSE) 2023 and received multiple research and higher-study opportunities from IITs and NITs
- Branch Topper in M.Tech (CSE-AIML) first semester with 9.32 SGPA
- Branch Topper in B.Tech CSE during 3rd, 5th, and 6th semesters
- Cleared internal selection round of Smart India Hackathon (SIH)
- Top-3 Finalist at Nirman 1.0 Robotics Competition (2022)